Vacuum Cleaner Problem

Aim:

To write a Python program to simulate a vacuum cleaner agent that cleans dirty rooms based on their states.

Algorithm:

1. Start the program.
2. Initialize the environment and the condition of each room.
3. Position the vacuum cleaner in a room.
4. Check whether the current room is dirty.
5. If the room is dirty, clean it.
6. If the room is clean, move to the next room according to the defined rules.
7. Repeat until all rooms are clean.
8. Display the final state of the environment.
9. Stop the program.

Code :

exp06_vacuum_cleaner.py

```python
class VacuumCleanerAgent:
    def __init__(self, initial_location='A', room_states=None):
        self.location = initial_location
        self.room_states = room_states if room_states else {'A': 'Dirty', 'B': 'Dirty'}
        self.cost = 0

    def sense(self):
        return self.location, self.room_states[self.location]

    def act(self):
        loc, status = self.sense()
        print(f"Agent at Location {loc} | Status: {status}")

        if status == 'Dirty':
            print(f"Action: SUCK dirt at Location {loc}")
            self.room_states[loc] = 'Clean'
            self.cost += 1
        else:
            new_loc = 'B' if loc == 'A' else 'A'
            print(f"Action: MOVE to Location {new_loc}")
            self.location = new_loc
            self.cost += 1

    def run(self):
        print(f"Initial State: {self.room_states}, Agent Start: {self.location}\n")
        steps = 0
        while any(status == 'Dirty' for status in self.room_states.values()) and steps < 6:
            self.act()
            steps += 1
            print(f"Current State: {self.room_states}\n")

        print("Final Result:")
        print(f"All rooms cleaned! Total actions taken: {self.cost}")

agent = VacuumCleanerAgent(initial_location='A', room_states={'A': 'Dirty', 'B': 'Dirty'})
agent.run()
```

Output :

```
PS D:\CSA1702-AI\Experiments> & C:\python311\python.exe d:/CSA1702-AI/Experiments/exp06_vacuum_cleaner.py
Initial State: {'A': 'Dirty', 'B': 'Dirty'}, Agent Start: A

Agent at Location A | Status: Dirty
Action: SUCK dirt at Location A
Current State: {'A': 'Clean', 'B': 'Dirty'}

Agent at Location A | Status: Clean
Action: MOVE to Location B
Current State: {'A': 'Clean', 'B': 'Dirty'}

Agent at Location B | Status: Dirty
Action: SUCK dirt at Location B
Current State: {'A': 'Clean', 'B': 'Clean'}

Final Result:
All rooms cleaned! Total actions taken: 3
PS D:\CSA1702-AI\Experiments>
```

Result :

Thus, the Python program to simulate the Vacuum Cleaner Problem was executed successfully.